Autonomous system and Routing

Shortest path routing: find least cost from source to destination

- min cost spanning tree

- hop count
- link bandwidth
- delay
- utilization

Collect info → process it → routing table

Distance Vector Routing:

vector of distances (hop count)

Who to share with
- neighbors

What to share:
- hop count table

distance, destination

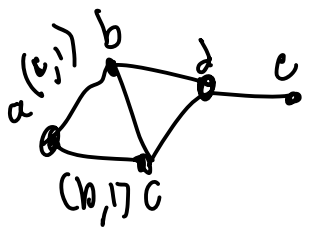
Each router has one of these

A's routing table

Dest Subnet	Hop Count	Next Hop
B	1	B
C	1	C
D		
E		

B's routing table

Dest Subnet	Hop Count	Next Hop
A	1	A
C	1	C
D	1	D
E	2	D



Bellman-Ford

Each router has a local view of the network

- doesn't tell you who anyone else is connected to
- doesn't know how it gets their

RIP2 → two labs from now (Router Information Protocol)

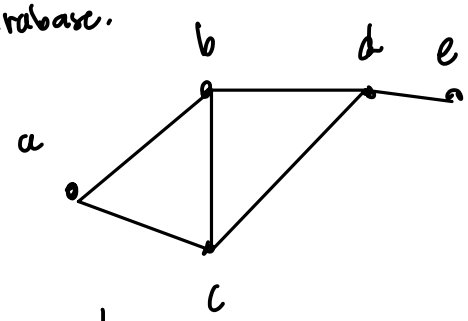
Issues:

- Scalability - network stabilization takes longer
- count to infinity problems (loops)

$\infty = 16$ → if you count to 16, unreachable

Link State Routing: weighted adjacency matrix

Generate a link state database, which is a weighted adjacency matrix but is called the link state database.



Who to share with:

Link state Database

What to share:

	A	B	C	D	E		
A		2	1				
B	3		2	2			
C	2	2		1			
D		1	1		4		
E				3			

OSPF - Dijkstra's Algorithm

- lots of features
- hard to implement

RIPv2 → Routing Information Protocol V2

Interior gateway protocol (so within an autonomous system)

shortest path calculation

Link State

Dijkstras shortest path algorithm

Features:

Multiple distance metrics

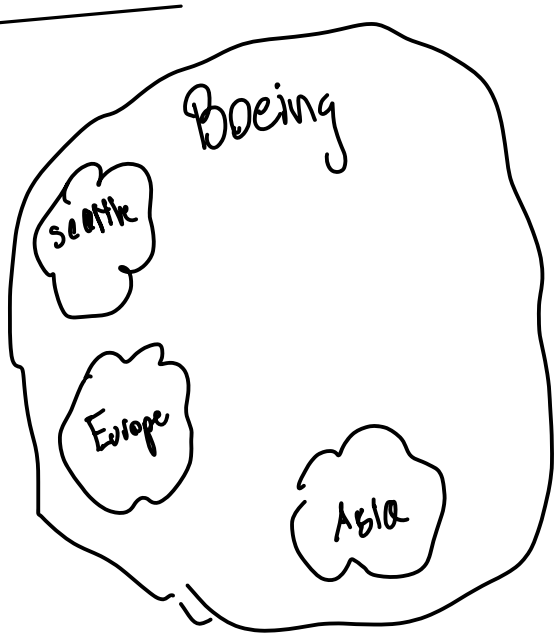
Dynamic

QoS

Load Balancing

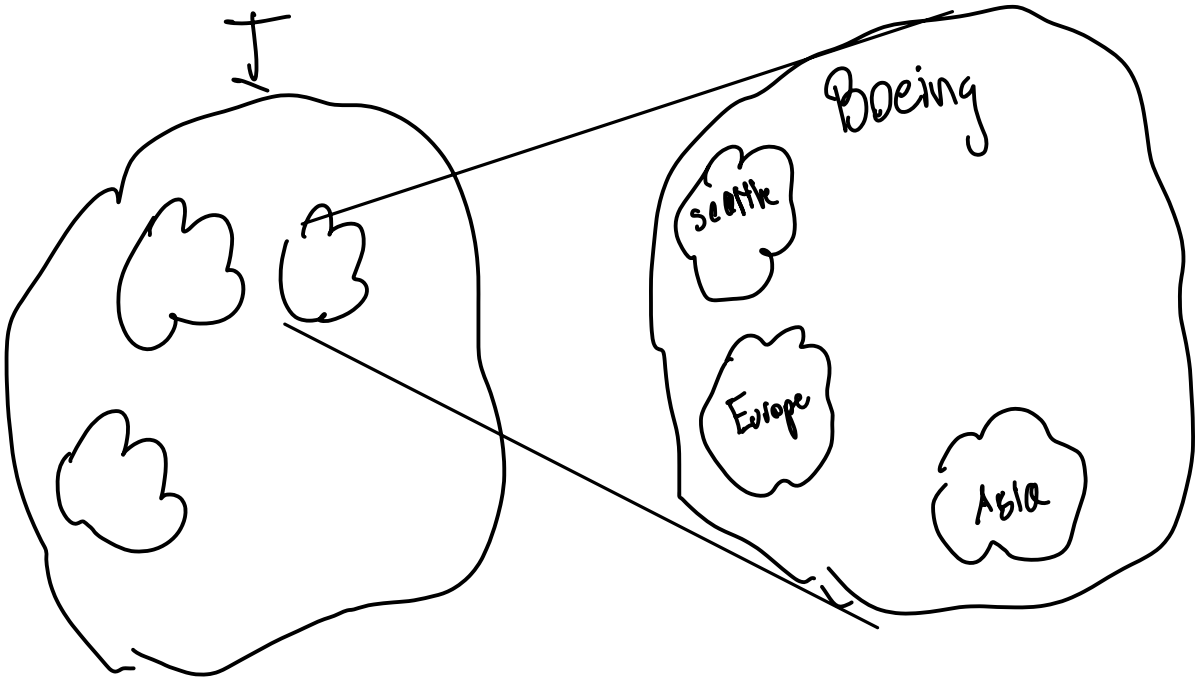
Hierarchical (support areas)

OSPF & Areas



Areas:

Break up OSPF cloud into smaller segments, within that we learn about the routing



Interior Gateway Protocol:

Interior Routing:

Sharing routes within a AS or company network

- Routing protocols : OSPF, IS-IS, RIPv2, EIGRP

Exterior Gateway Protocol: How do autonomous systems share information with eachother

- sharing between AS
-

BGP - Border Gateway Protocol

Exterior Border Gateway Protocol
- speaks BGP between ASes

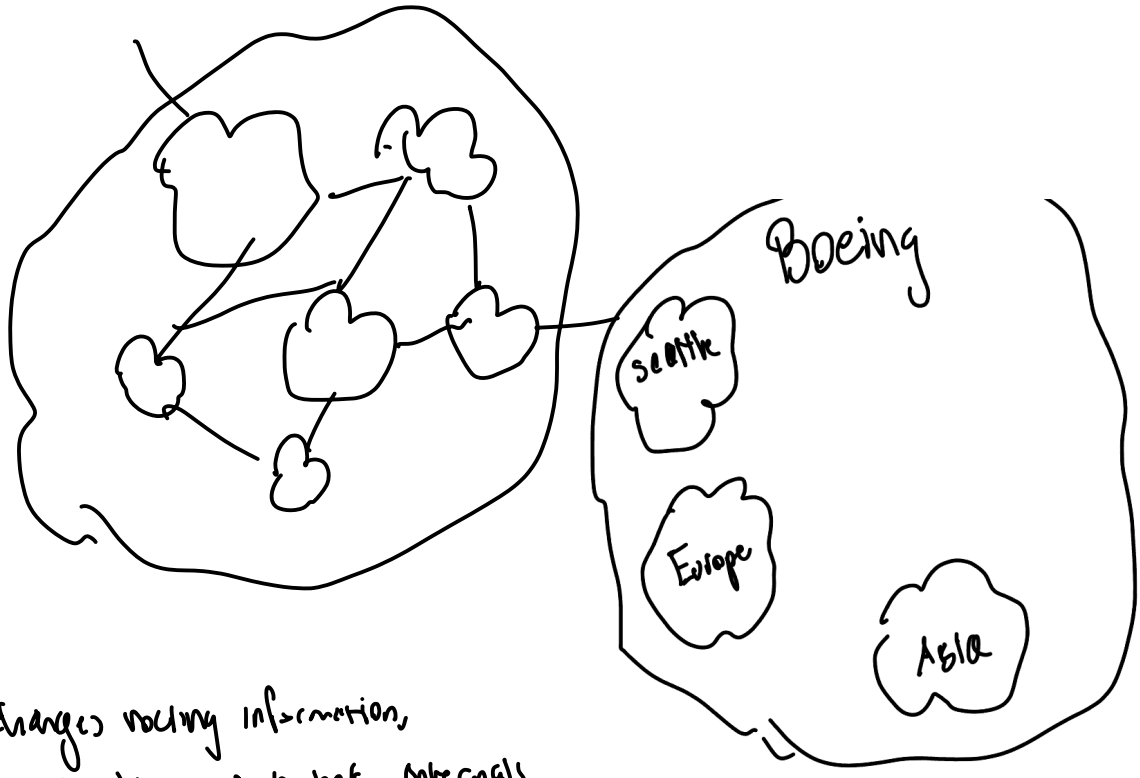
IBGP - interior BGP
- speaks BGP with other BGP speakers in the AS

Used to route between ASes

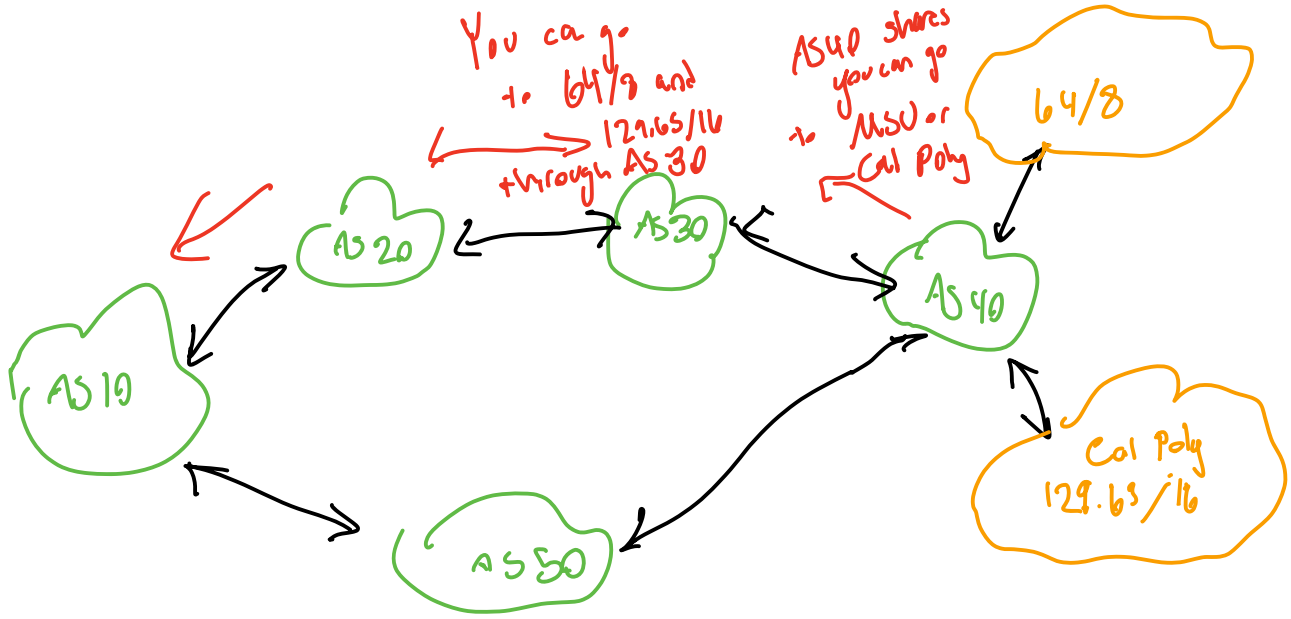
- exchanges network reachability information
- says you can get to network X via this path of ASes

- Needs to be concerned with policies (business & political)
  - uses a modified Distance Vector Protocol
    - Path-Vector Routing

Runs over reliable transport protocol (TCP)



BGP - exchanges routing information, reachability, but not internals



What defines the boundary of the Autonomous system?

- Who owns what routers

edge routers are the edge.

CEF → optimization of RIB

Routing RIB and FIB

RIB = Router Information Base

- stores all learned routes
- selects the best path
- feeds into FIB process
- control plane

how they update tables

FIB = Forwarding Information Base

searched at 10Gbps

contains only best routes

- used to actually find path for forwarding packets
- data plane

what a packet uses to get data out

TCAM → tertiary content addressable memory

Content Addressable Memory

- 0 or 1 value

129.65/16

ICMP - ping packets (IP <> IP)

Internet Control Message Protocol

- RFC 792
- provides communication between IP on one machine and IP on another.